

India: New nuclear rules will boost domestic reactors

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India

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New nuclear rules will boost domestic reactors over foreign alternatives

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Short-term trajectory:

Long-term trajectory:

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New nuclear sector legislation is likely to spur increased demand for pressurized heavy-water reactors (PHWRs) developed domestically, largely driven by Indian heavy industry and power sector demand.

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Future demand for small modular reactors (SMRs) will be determined by AI-related infrastructure like data centers but will face regulatory barriers.

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The final rules are likely to integrate nuclear power more closely into India's green finance frameworks.

Government expected to enact the final rules and regulations of its nuclear power overhaul by year-end. Draft rules under the Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Act were released on 18 August, and public feedback closed on 4 September.

Parliament passed the act last December, but the subsidiary rules needed to implement it remain pending. The power industry expects these to be finalized in the next few months, with licensing and technical norms to follow early next year. That timeline would also give the government room to amend the SHANTI Act during parliament's budget session in February 2027.

The act opens the nuclear power sector to limited private sector participation. Private companies can commission both domestic and foreign firms to build nuclear reactors and use the power generated, but the reactors would remain operationally under the control of the state-owned Nuclear Power Corporation of India. The government would retain tight regulatory control of all elements of the nuclear fuel cycle (see: Opening of nuclear power sector will depend on bureaucratic buy-in , 9 January 2026).

SHANTI Act will allow India to harness private capital to expand its nuclear capacity. India has targeted 100 gigawatts (GW) of nuclear power by 2047, roughly a twelvefold increase of its present installed capacity of 8 GW. The decision to open the sector followed 15 years of stunted growth under the 2010 Civil Liability for Nuclear Damage Act, whose flawed provisions made insurance for nuclear component makers nearly impossible. The Department of Atomic Energy (DAE) had long resisted the idea of foreign players and technologies entering the sector, a stance shaped by decades of Western sanctions. The SHANTI Act indicates Prime Minister Narendra Modi's government has largely overcome that resistance,

though the DAE retains a high degree of control over the introduction of new technologies. While the government's targets are overly ambitious, private investment will give a big boost to new reactor construction. Even before the changes take effect, nearly a third of India's new private capital expenditure in the first half of this year was nuclear-driven.

New demand from heavy industry and utilities will be for domestically produced PHWR reactors. The final rules are likely to steer initial private investment toward the domestic 700 megawatt (MW) PHWR. The SHANTI Act places strong emphasis on plants' financial security. Financial cover and insurance must span the reactor's entire life cycle, including decommissioning and spent fuel clean-up. This scope is reflected in the proposed composite nuclear license, which covers construction, operation, and decommissioning under a single instrument. In addition, the draft rules propose reviewing operators' maximum liability every five years.

All of these provisions will bias insurance and lending towards the domestic PHWRs, given their safety record, established supply ecosystem, and low production and operation costs. Their 90%-95% local content adds further appeal at a time when geopolitics has weaponized supply chains and sanctions have led to surges in the prices of imported components. Initial interest in commissioning reactors has come entirely from local power companies and heavy industry, including steel and aluminum.

SMR development and data center demand will be the key determinants of foreign involvement. PHWRs use minimal foreign components, though some private owners may seek bespoke elements requiring foreign technology. The market is likely to open for foreign firms once functional SMRs become available. However, the DAE is likely to create regulatory hurdles for technological innovation it fears will reduce its standing as the country's preeminent nuclear development body. Already, the new rules permit new reactor types only if they have a proven operating track record overseas. Indian demand for such reactors is expected to come chiefly from data centers feeding the AI infrastructure plans of large US tech firms like Google and Amazon, whose creditworthiness should limit counterparty risk.

Even though there has been little interest in the DAE-designed 220 MW PHWR, partly because no unit has been built since 2011, the DAE has not grown more receptive to foreign alternatives. Larger foreign reactors are unlikely to make inroads without substantial support from overseas export credit agencies and will run into even greater DAE resistance.

The new rules will partially embed nuclear power into India's green financial frameworks. The draft rules left nuclear power's place in India's green finance framework unresolved. The government is expected to clarify that nuclear power has infrastructure status, enabling lower-rate bank lending and tax concessions for investors. The finance ministry will probably allow nuclear power to obtain financing from sovereign green bond issuances once liability cover is settled. Industry calls for viability gap funding are unlikely to win support from the government unless a setback, such as a stalled SMR tender or financing shortfall, makes government intervention unavoidable.

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